

Physical Chemistry (I)

Lecture 08

Entropy



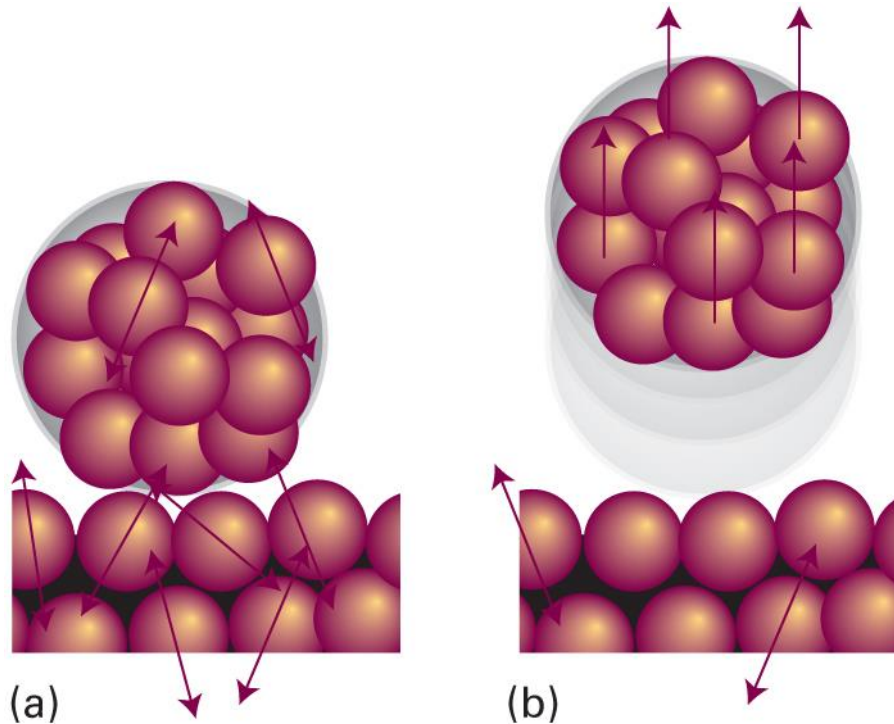
In the Last Class

- Heat engines
 - Arrow of time
 - Second law of thermodynamics
 - Carnot cycle
- Entropy
 - Thermodynamic definition
 - Clausius inequality
 - Fundamental equation



Last class

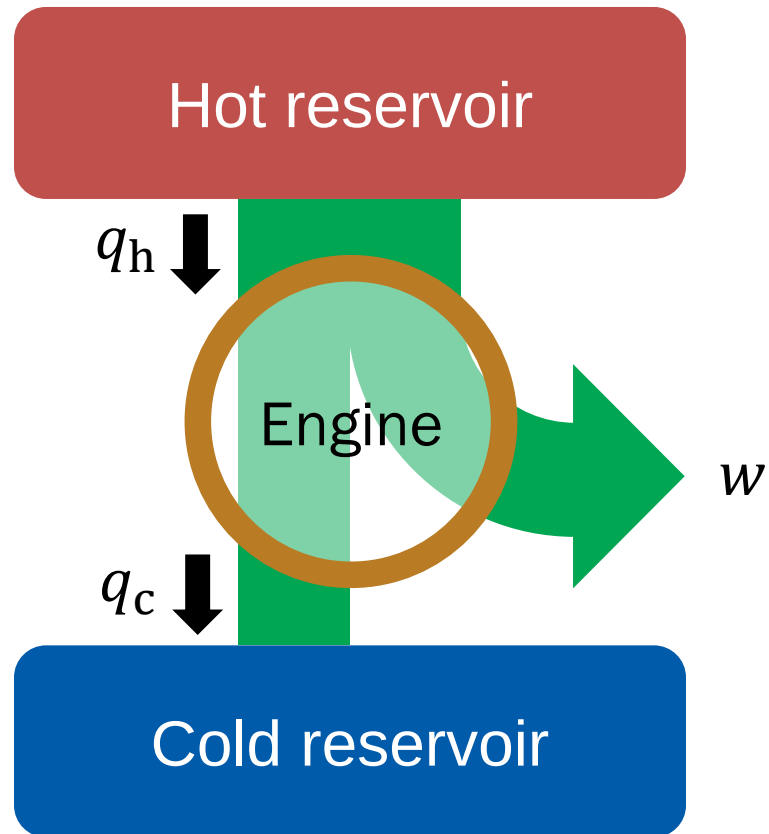
Irreversibility: Heat and Work



Heat and work are not equal

Last class

Heat Engine



$$q_h + q_c + w = 0$$

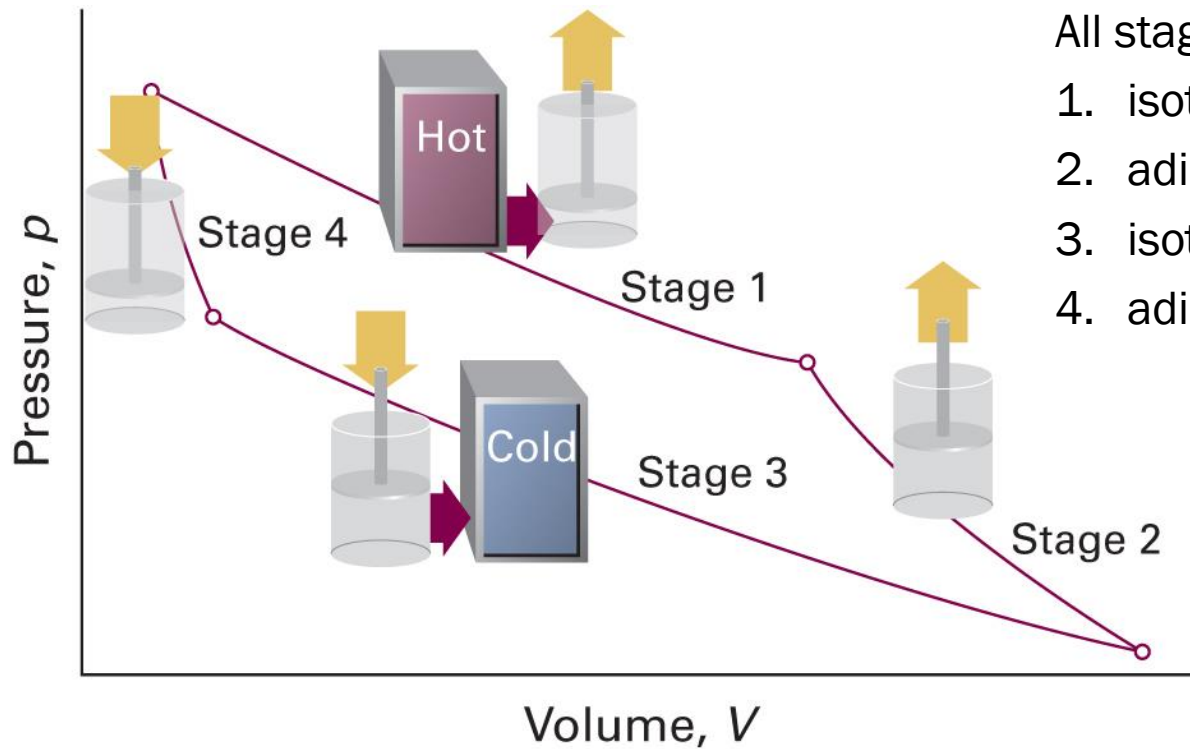
Last class

Second Law of Thermodynamics

- **Kelvin:** “No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.”
- **Clausius:** “Heat does not flow spontaneously from a cool body to a hotter body.”



Carnot Cycle



- All stages are reversible;
1. isothermal expansion
 2. adiabatic expansion
 3. isothermal compression
 4. adiabatic compression

Entropy Calculation

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- Entropy change

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

- Molar entropy change $\Delta S_{\text{m}} = \Delta S/n$

Clausius Inequality

- Clausius inequality

$$dS \geq \frac{\delta q}{T}$$

- *3rd version* of the second law of thermodynamics:
“The entropy of an isolated system never decreases.”

Fundamental Equation

$$dU = TdS - pdV$$

As $U = U(S, V)$ is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S} \right)_V$$

$$p = - \left(\frac{\partial U}{\partial V} \right)_S$$

Last class

Examples: Entropy Calculation

Heat transfer

Expansion (reversible vs. free)

Phase transitions

Heating



Entropy Calculation

- Heat transfer to surroundings can be always considered as reversible heat transfer

$$\delta q_{\text{sur}} = \delta q_{\text{sur,rev}} \text{ (always)}$$

- Reversible change:

$$dS = \delta q_{\text{rev}}/T = \delta q/T$$

$$dS_{\text{sur}} = \delta q_{\text{sur,rev}}/T = -\delta q/T$$

- Irreversible change:

$$dS \neq \delta q/T \text{ and } dS_{\text{sur}} = -\delta q/T$$

Reversible Expansion

$$\delta q = -\delta q_{\text{sur}}$$

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = -\frac{q_{\text{rev}}}{T} = -\Delta S$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} = 0$$

Free Expansion

For a perfect gas, isothermal expansion accompanies with

$$\Delta S = nR \ln \left(\frac{V_f}{V_i} \right)$$

However, $w = 0$ (free expansion) and $\Delta U = 0$ (perfect gas)

$$q = q_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = 0$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} > 0$$

Reversible Phase Transition

Normal transition temperature T_{trs} : the temperature at which two phases are in equilibrium at 1 atm

If the system is at T_{trs} and 1 atm,

$$q = \Delta_{\text{trs}}H \text{ (constant pressure)}$$

$$\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{sur}} = -\Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{total}} = \Delta_{\text{trs}}S + \Delta_{\text{trs}}S_{\text{sur}} = 0$$

Heating at Constant V

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

At constant volume, $\delta q_{\text{rev}} = dU = C_V dT$

$$S(T_f) - S(T_i) = \int_{T_i}^{T_f} \frac{C_V}{T} dT$$

For constant C_V ,

$$S(T_f) = S(T_i) + C_V \int_{T_i}^{T_f} \frac{1}{T} dT = S(T_i) + C_V \ln \left(\frac{T_f}{T_i} \right)$$

Heating at Constant p

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

At constant pressure, $\delta q_{\text{rev}} = dH = C_p dT$

$$S(T_f) - S(T_i) = \int_{T_i}^{T_f} \frac{C_p}{T} dT$$

For constant C_p ,

$$S(T_f) = S(T_i) + C_p \int_{T_i}^{T_f} \frac{1}{T} dT = S(T_i) + C_p \ln \left(\frac{T_f}{T_i} \right)$$

Nernst Heat Theorem

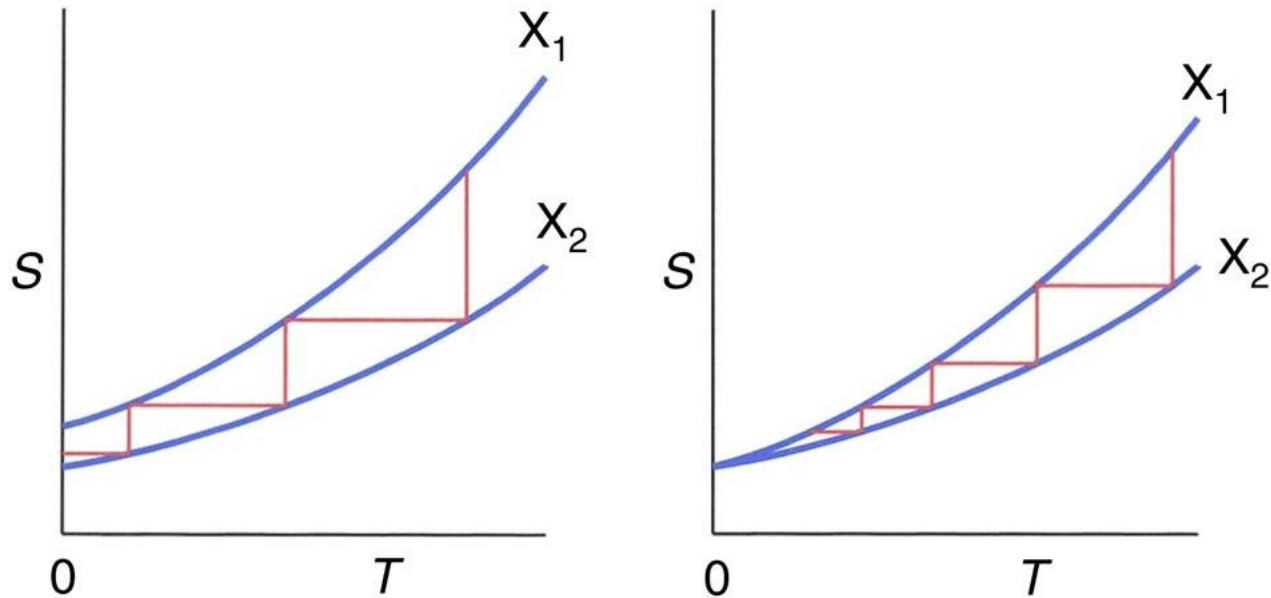
“The entropy change accompanying any transformation approaches zero as the temperature approaches zero:

$$\Delta S \rightarrow 0 \text{ as } T \rightarrow 0,$$

if all the substances involved are perfectly ordered.”

Not in Atkins

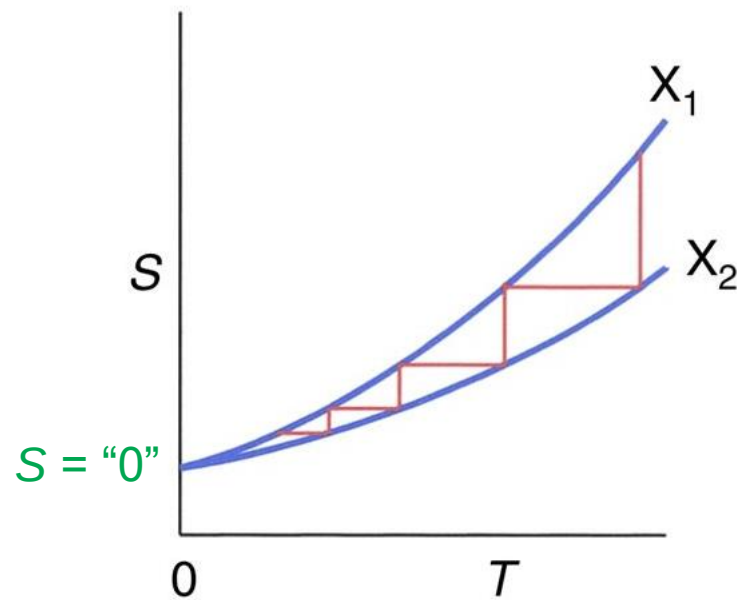
Unattainability Principle



(Image: Masanes and Oppenheim, *Nature Communications* 8: 14538 (2017), Figure 1)

Third Law of Thermodynamics

“The entropy of all perfect crystalline substances is zero at $T = 0$.”



(Image: Masanes and Oppenheim, *Nature Communications* 8: 14538 (2017), Figure 1)

Third-Law Entropy

$$S(T) - \underbrace{S(0)}_{=0} = \int_0^T \frac{C_p(T')}{T'} dT'$$

$$S(T) = \int_0^T \frac{C_p(T')}{T'} dT'$$

Debye T^3 Law

For a crystalline solid at $T \sim 0$,

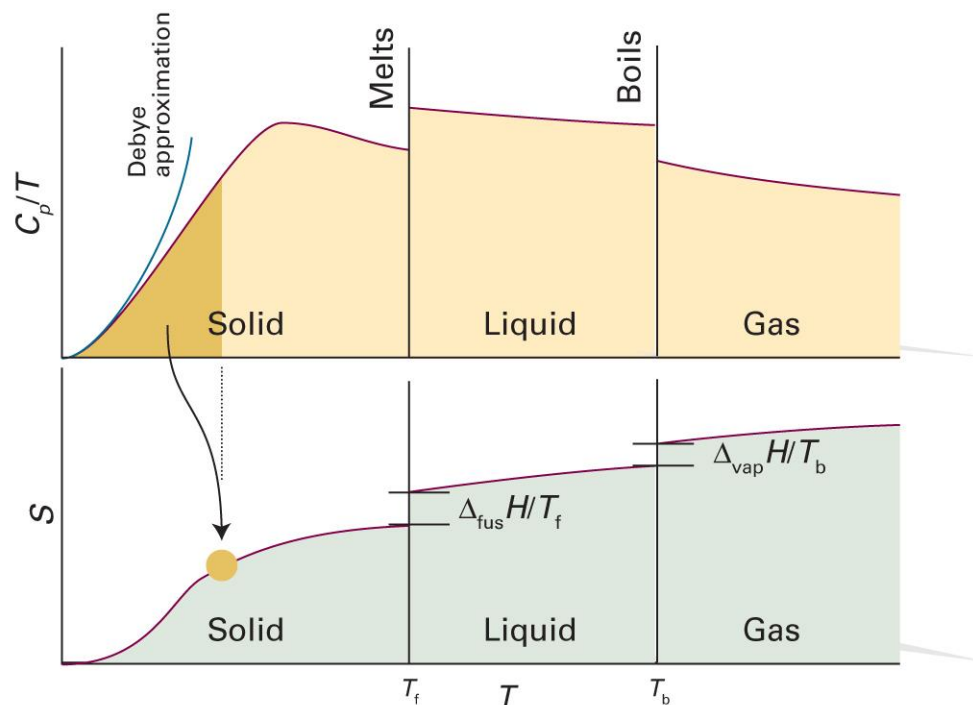
$$C_{p,m}(T) = aT^3$$

Hence,

$$S_m(T) = \int_0^T \frac{C_{p,m}(T')}{T'} dT' = \int_0^T \frac{a(T')^3}{T'} dT' = \frac{1}{3} aT^3 = \frac{C_{p,m}}{3}$$

when T is close to zero.

Typical Entropy Curve



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 3C.1)

Thermochemistry of Entropy

- Standard entropy $S^\ominus$: third-law entropy of a pure substance at 1 bar (always positive)
- Molar standard entropy $S_m^\ominus = S^\ominus/n$
- Standard reaction entropy $\Delta_r S^\ominus$

$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

Enthalpy vs. Entropy

- **Enthalpy:** no natural absolute value

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f H^\ominus(j)$$

- **Entropy:** third-law absolute value

$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

One Exception

$$S^{\ominus}(\text{H}^+, \text{aq}) = 0: \text{convention}$$

The entropies of ions in water are values
relative to the hydrogen ion in water
→ can either be positive or negative

Temperature Dependence

$$S_m(T_2) = S_m(T_1) + \int_{T_1}^{T_2} \frac{C_{p,m}(T)}{T} dT$$

Using the definition of standard reaction entropy,

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \int_{T_1}^{T_2} \frac{\Delta_r C_p^\ominus(T)}{T} dT$$

Similar to Kirchhoff's law!

Temperature Dependence

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \int_{T_1}^{T_2} \frac{\Delta_r C_p^\ominus(T)}{T} dT$$

If $\Delta_r C_p^\ominus$ is constant,

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \Delta_r C_p^\ominus \int_{T_1}^{T_2} \frac{1}{T} dT$$

$$\Delta_r S^\ominus(T_2) = \Delta_r S^\ominus(T_1) + \Delta_r C_p^\ominus \ln(T_2/T_1)$$

So, what does entropy mean?

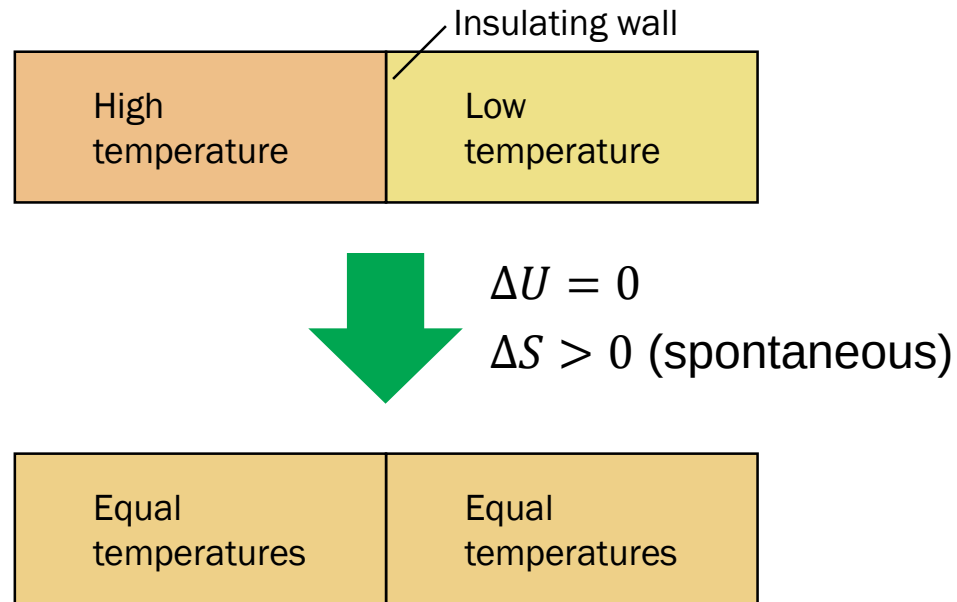
Second Law

$$dS \geq 0 \text{ (for an isolated system)}$$

- $dS = 0$: reversible change, equilibrium
- $dS > 0$: irreversible change, spontaneous change

Not in Atkins

Entropy and Work



Loss of capacity to do work

Index of Exhaustion

Entropy can be considered as an **index of exhaustion**

- Clausius: “The energy of the universe is constant; the entropy tends to a maximum.”
- Heat death of the universe: when S_{universe} is maximized
- Jeremy Rifkin, *Entropy: A New World View* (1980)

Statistical Consideration

- (Macro)state: defined by values of **macroscopic** variables
- Microstate: specific **microscopic** arrangement
- Different microstates can be mapped into the same macrostate

Example: Coin Toss

When tossing three coins,

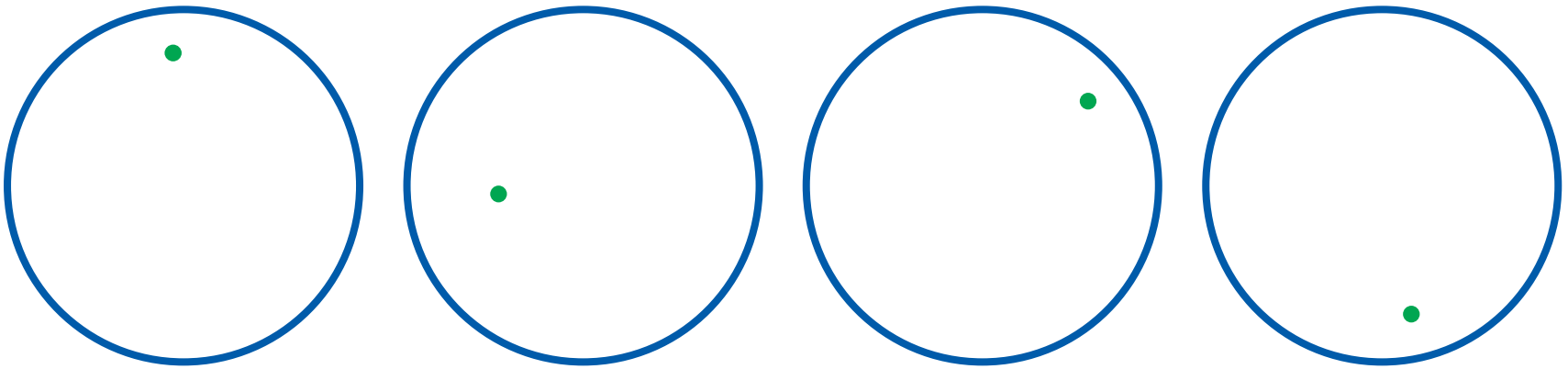
- **Macrostate:** defined by the number of heads h
- **Microstate:** the actual arrangement of coins



Example: Diffusion

When a gas molecule moves around the system,

- **Macrostate:** defined by the volume V
- **Microstate:** the position of the molecule



Number of Microstates W

“The number of microstates
corresponding to a specific macrostate”

- Coin toss

$$W(h = 0) = 1, W(h = 1) = 3, W(h = 2) = 3, W(h = 3) = 1$$

- Diffusion

Let the volume occupied by a molecule v_0 ;

for the system volume V , $W(V) = V/v_0$

Boltzmann's Formula

$$S = k_B \ln W$$

number of microstates

Boltzmann constant (1.38×10^{-23} J/K)

$$\Delta S = S_f - S_i = k_B \ln W_f - k_B \ln W_i = k_B \ln \left(\frac{W_f}{W_i} \right)$$

Example: Diffusion

Suppose that the system volume changes from V_i to V_f :

$$W_i = V_i/v_0$$

$$W_f = V_f/v_0$$

The change in entropy is given by

$$\Delta S_1 = k_B \ln(W_f/W_i) = k_B \ln(V_f/V_i)$$

For a perfect gas consisting of N molecules,

$$\Delta S_N = Nk_B \ln(V_f/V_i) = nR \ln(V_f/V_i)$$

Second Law Revisited

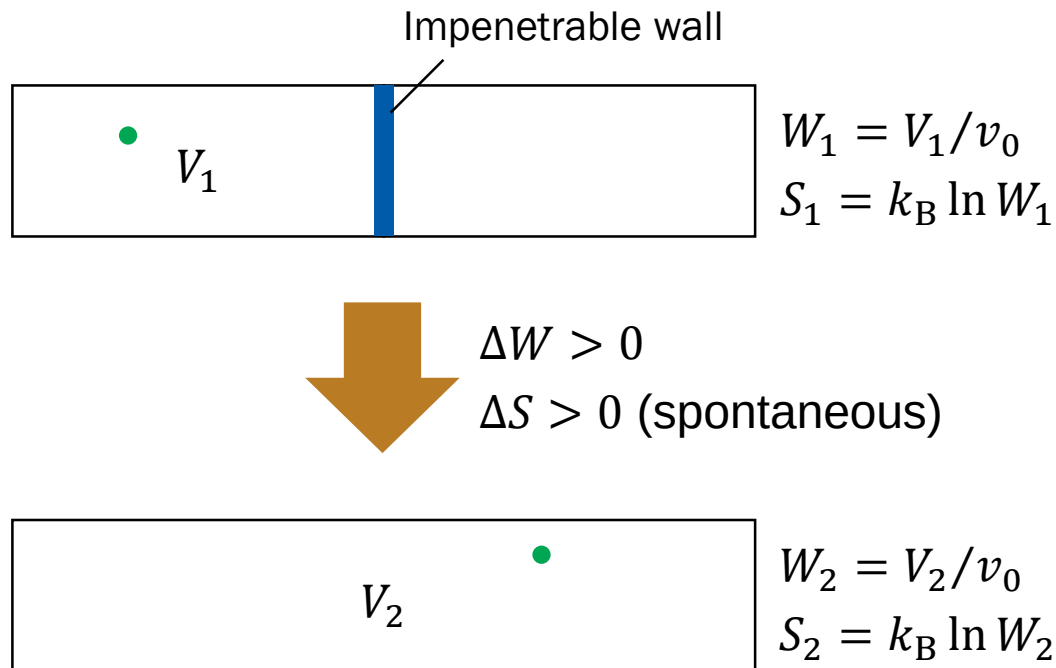
$$dS \geq 0 \text{ (for an isolated system)}$$

As $S = k_B \ln W$ is a monotonic function of W ,

$$dW \geq 0 \text{ (for an isolated system)}$$

The number of microstates never decreases

Example: Diffusion



Third Law Revisited

At $T = 0$,

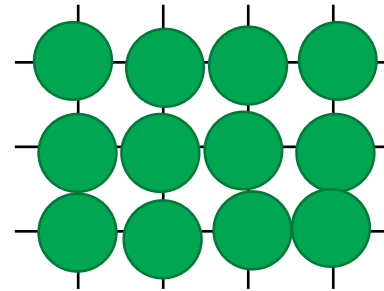
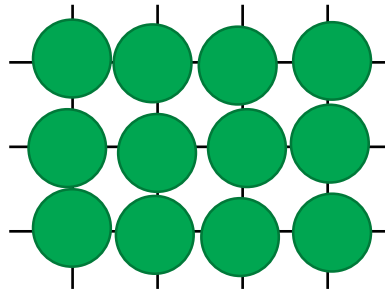
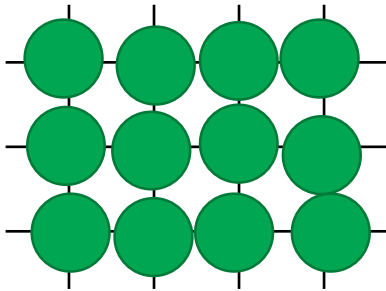
$$S = k_B \ln W = 0$$

$$W = 1$$

Only one microstate is possible at 0 K

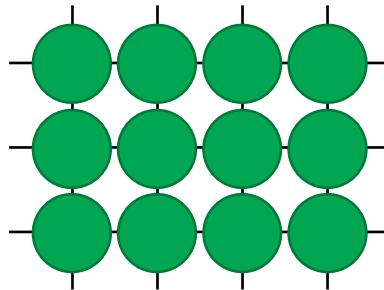
Solid at Absolute Zero

$$T > 0$$



...

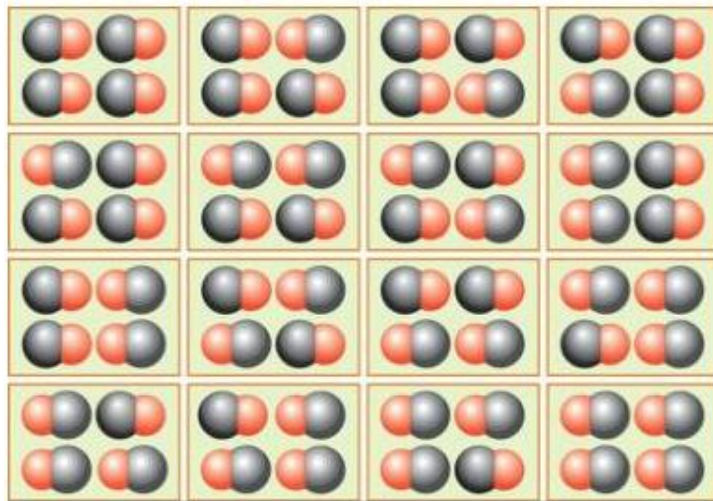
$$T = 0$$



Residual Entropy

$W > 1$ and $S > 0$ even at $T = 0$

e.g. carbon monoxide (CO)

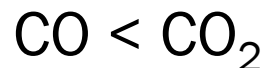


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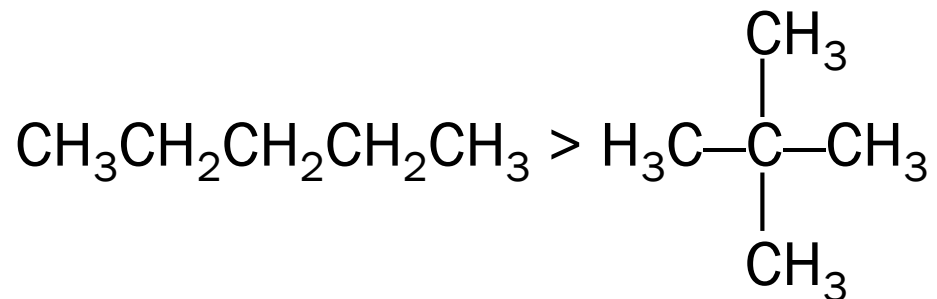
Molecular View on Entropy

In general, W is larger (and S is greater) for a molecule with

1. more atoms



2. more flexibility



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Polymer Conformations



$$\Delta W > 0$$

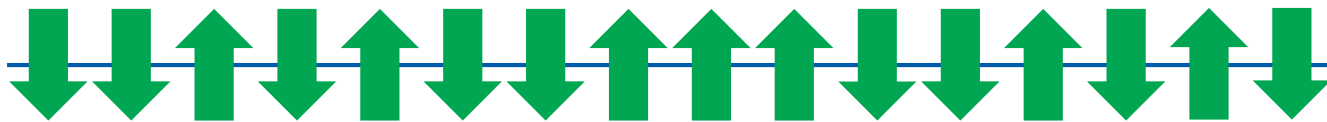
$$\Delta S > 0 \text{ (spontaneous)}$$



Spins

Electrons and protons have spins: up ($\uparrow$) or down ($\downarrow$)

Consider a solid system consisting of N spins:



- Two spins: 4 microstates ($\uparrow\uparrow$, $\uparrow\downarrow$, $\downarrow\uparrow$, $\downarrow\downarrow$)
- Three spins: 8 microstates
- N spins: 2^N microstates

Similar!



Spins in a Magnetic Field

The two spins react to a magnetic field differently:

$$E(\uparrow) = +\frac{\epsilon}{2}, \quad E(\downarrow) = -\frac{\epsilon}{2}$$

The internal energy of the system is

$$U = \frac{\epsilon}{2}N_{\uparrow} - \frac{\epsilon}{2}N_{\downarrow} = \frac{\epsilon}{2}(N_{\uparrow} - N_{\downarrow})$$

Macrostate

- Total number of spins $N = N_{\uparrow} + N_{\downarrow}$
- Internal energy

$$U = \frac{\epsilon}{2} (N_{\uparrow} - N_{\downarrow}) = \epsilon N_{\uparrow} - \frac{\epsilon}{2} N$$

- **Macrostate:** number of “up” spins

Entropy

For a given macrostate (number of up spins = M),
what is the number of microstates?

M spins are “up” among N spins: combinations

$$W = {}_N C_M = \frac{N!}{M! (N - M)!}$$

$$S = k_B \ln W = k_B \ln \left[\frac{N!}{M! (N - M)!} \right]$$

Stirling's Approximation

$$\ln N! \approx N \ln N - N \text{ (for large } N\text{)}$$

$$\begin{aligned}\ln N! &= \ln\{N \times (N-1) \times (N-2) \times \cdots\} \\ &= \ln N + \ln(N-1) + \ln(N-2) + \cdots \\ &= \sum_{k=1}^N \ln k \\ &\approx \int_1^N \ln x \, dx \\ &= [x \ln x - x]_1^N \\ &= N \ln N - N + 1 \\ &\approx N \ln N - N \text{ (for large } N \gg 1\text{)}\end{aligned}$$

Simplification

$$\begin{aligned} S &= k_B \ln \left[\frac{N!}{M! (N - M)!} \right] \\ &= k_B [\ln N! - \ln M! - \ln(N - M)!] \end{aligned}$$

Using Stirling's approximation,

$$S = k_B [N \ln N - M \ln M - (N - M) \ln(N - M)]$$

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Entropy and Internal Energy

$$S = k_B [N \ln N - M \ln M - (N - M) \ln(N - M)]$$

For a system with M up spins,

$$U = \epsilon M - \frac{\epsilon}{2} N$$

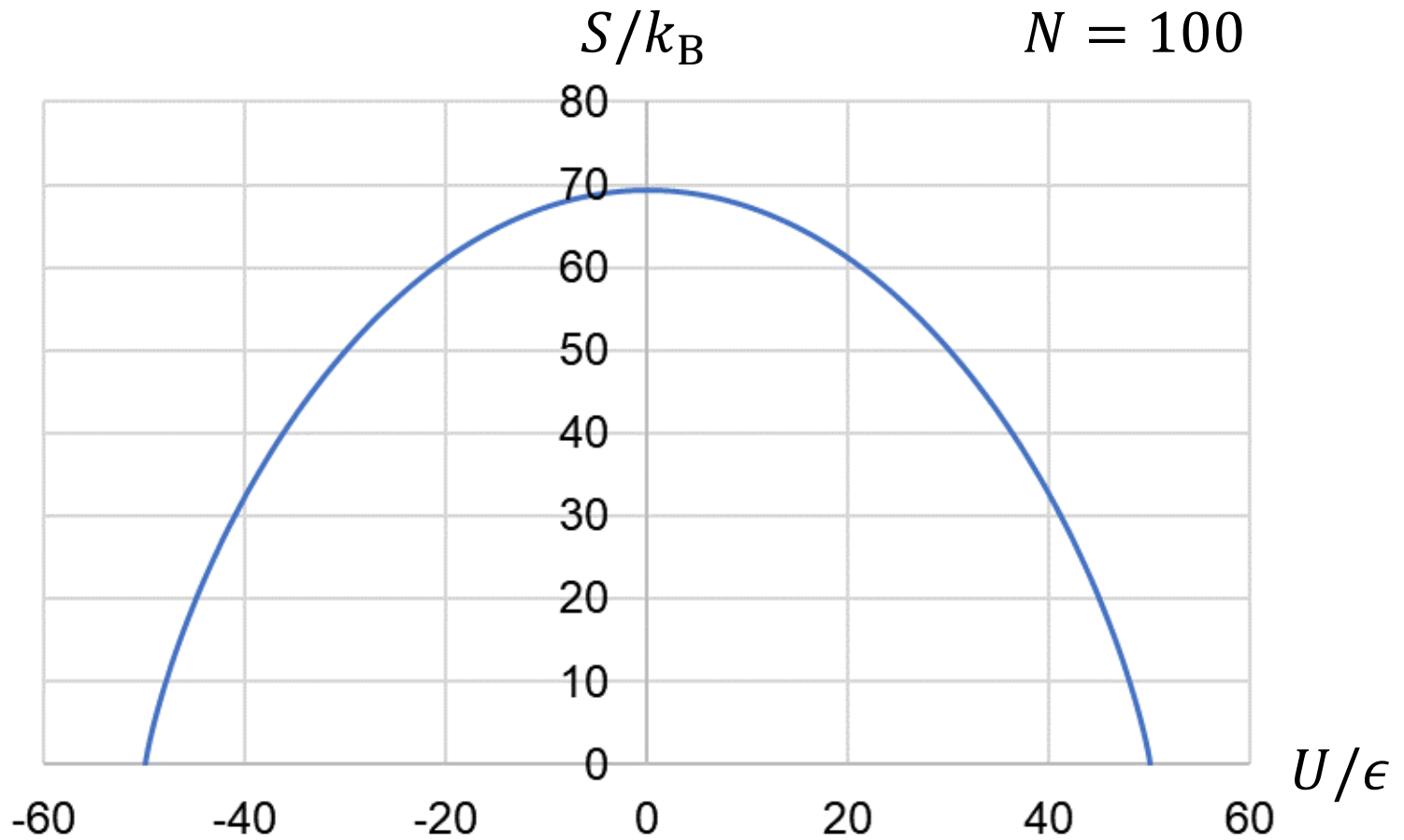
$$M = \frac{N}{2} + \frac{U}{\epsilon}$$

$$S = k_B [N \ln N$$

$$- \left(\frac{N}{2} + \frac{U}{\epsilon} \right) \ln \left(\frac{N}{2} + \frac{U}{\epsilon} \right) - \left(\frac{N}{2} - \frac{U}{\epsilon} \right) \ln \left(\frac{N}{2} - \frac{U}{\epsilon} \right)]$$

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$$S(U)$$



Temperature

From the fundamental equation of thermodynamics,

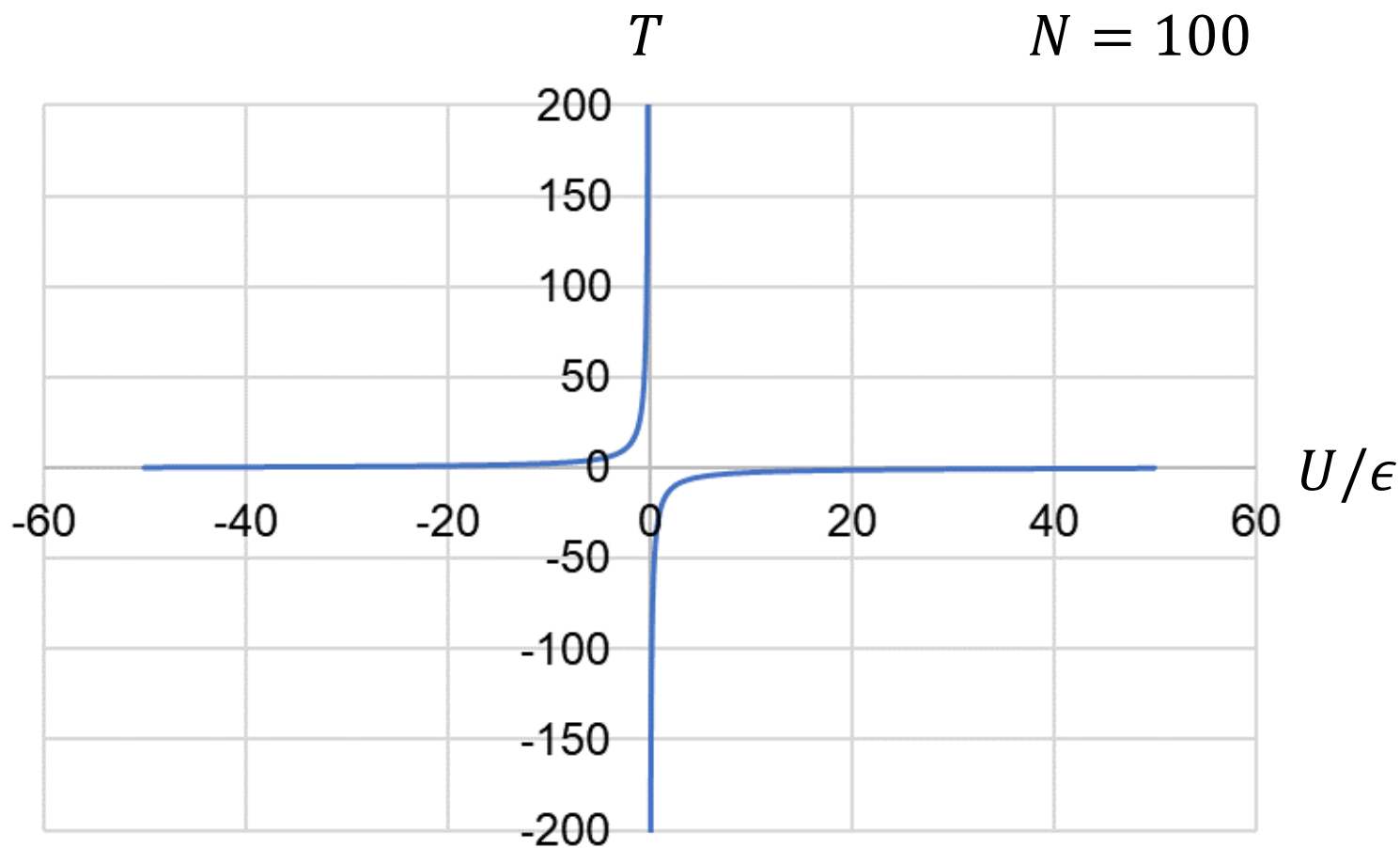
$$T = \left(\frac{\partial U}{\partial S} \right)_V, \quad \frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_V$$

$$S = k_B \left[N \ln N - \left(\frac{N}{2} + \frac{U}{\epsilon} \right) \ln \left(\frac{N}{2} + \frac{U}{\epsilon} \right) - \left(\frac{N}{2} - \frac{U}{\epsilon} \right) \ln \left(\frac{N}{2} - \frac{U}{\epsilon} \right) \right]$$

$$\frac{1}{T} = \frac{k_B}{\epsilon} \ln \left(\frac{N}{2} - \frac{U}{\epsilon} \right) - \frac{k_B}{\epsilon} \ln \left(\frac{N}{2} + \frac{U}{\epsilon} \right) = \frac{k_B}{\epsilon} \ln \left(\frac{N/2 - U/\epsilon}{N/2 + U/\epsilon} \right)$$

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$$T(U)$$



Meaning of Temperature

$$\frac{1}{T} = \frac{k_B}{\epsilon} \ln \left(\frac{N/2 - U/\epsilon}{N/2 + U/\epsilon} \right)$$

- Number of up spins: $N_{\uparrow} = M = N/2 + U/\epsilon$
- Number of down spins: $N_{\downarrow} = N - M = N/2 - U/\epsilon$

$$\frac{1}{T} = \frac{k_B}{\epsilon} \ln \left(\frac{N_{\downarrow}}{N_{\uparrow}} \right)$$

- If $N_{\downarrow} > N_{\uparrow}$, the temperature is **positive**
- If $N_{\downarrow} < N_{\uparrow}$, the temperature is **negative**

Measuring Negative T

PERSPECTIVES

PHYSICS

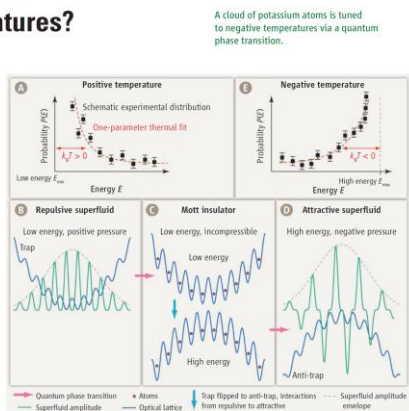
Negative Temperatures?

Lincoln D. Carr

Ultracold quantum gases present an exquisitely tunable quantum system. Applications include precision measurement (1), quantum simulations for advanced materials design (2), and new regimes of chemistry (3). Typically trapped in a combination of magnetic fields and laser beams, strongly isolated from the environment in an ultrahigh vacuum, and cooled to temperatures less than a microdegree above absolute zero, they are the coldest known materials in the universe. The interactions between atoms in the gas can be tuned over seven orders of magnitude and from repulsive to attractive (4). The addition of standing waves made from interfering lasers at optical wavelengths gives rise to an optical lattice, a crystal of light, periodic just like the usual crystals made of matter. On page 52 of this issue, Braun *et al.* (5) use these special features of ultracold quantum gases to produce a thermodynamic oddity—negative temperature.

Temperature is casually associated with hot and cold. How can something be “colder” than absolute zero? The answer lies in a more precise notion of temperature. Temperature is a single-parameter curve fit to a probability distribution. Given a large number of particles, we can say each of them has a probability to have some energy, $P(E)$. Most will be in low-energy states and a few in higher-energy states. This probability distribution can be fit very well with an exponential falling away to zero. Of course, the actual distribution may be very noisy, but an exponential fit is still a good approximation (see the figure, panel A). Negative temperature means most particles are in a high-energy state, with a few in a low-energy state, so that the exponential rises instead of falls (see the figure, panel E).

To create negative temperature, Braun *et al.* had to produce an upper bound in energy, so particles could pile up in high-energy rather than low-energy states. In their experiment, there are three important kinds of energy: kinetic energy; or the energy of motion in the optical lattice; potential energy, due to magnetic fields trapping the gas; and interaction energy, due to interactions between the atoms in their gas. The lattice naturally gives an



Less than zero. (A) Temperature is a one-parameter fit: As the energy gets large, the probability that an atom will have that energy falls away exponentially. A quantum phase transition from a repulsive superfluid (B) to a Mott insulator (C) provides a bridge to an attractive superfluid (D), resulting in negative pressure balanced by negative temperature (E).

upper bound to kinetic energy via the formation of a band gap, a sort of energetic barrier to higher-energy states. The potential energy was made negative by the clever use of an anti-trap on top of the lattice, taking the shape of an upside-down parabola. Finally, the interactions were tuned to be attractive (negative). Thus, all three energies had an upper bound, and in principle, the atoms could pile up in high-energy states.

Braun *et al.* convinced their gas to undergo such a strange inversion using a quantum phase transition (6), an extension of the well-known thermodynamic concept of phase transitions to a regime in which the temperature is so low that it plays no role in the change of phase. In this case, they worked with two phases, superfluid and Mott insulator. In a superfluid, the gas flows freely without viscosity and is coherent, like a laser, but made of matter instead of light. In a Mott insulator, the atoms freeze into a regular pattern and become incompressible, similar to a solid.

A cloud of potassium atoms is tuned to negative temperatures via a quantum phase transition.

Braun *et al.* first make their atoms repulsive in a superfluid phase. They tune them to a Mott insulating phase by simply turning up the intensity of the optical lattice lasers, making the lattice deeper. Then they tune the interactions to be attractive and at the same time turn their trap upside down to be an anti-trap. Finally, they melt the Mott insulator to obtain an attractive superfluid. These anti-traps have been used before, to create a self-propagating pulse of atoms that does not disperse (a bright soliton) from attractive gases in one dimension (7, 8). However, attractive quantum gases in two and three dimensions can implode, rather spectacularly (9). This tendency to implode is called negative pressure. The negative temperature is precisely what stabilizes the gas against negative pressure and implosion; the Mott insulator serves as a bridge state between positive pressure and pressure, and negative temperature and pressure (see the figure, panels B to D).

Braun *et al.*'s exploration of negative

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Negative Absolute Temperature for Motional Degrees of Freedom

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Absolute temperature is usually bound to be positive. Under special conditions, however, negative temperatures—in which high-energy states are more occupied than low-energy states—are also possible. Such states have been demonstrated in localized systems with finite, discrete spectra. Here, we prepared a negative temperature state for motional degrees of freedom. By tailoring the Bose-Hubbard Hamiltonian, we created an attractively interacting ensemble of ultracold bosons at negative temperature that is stable against collapse for arbitrary atom numbers. The quasimomentum distribution develops sharp peaks at the upper band edge, revealing thermal equilibrium and bosonic coherence over several lattice sites. Negative temperatures imply negative pressures and open up new parameter regimes for cold atoms, enabling fundamentally new many-body states.

Absolute temperature T is one of the central concepts of statistical mechanics and is a measure of, for example, the amount of disorder motion in a classical ideal gas. Therefore, nothing can be colder than $T = 0$, where classical particles would be at rest. In a thermal state of such an ideal gas, the probability P_i for a particle to occupy a state i with kinetic energy E_i is proportional to the Boltzmann factor

$$P_i \propto e^{-E_i/k_B T} \quad (1)$$

where k_B is Boltzmann's constant. An ensemble at positive temperature is described by an occupation distribution that decreases exponentially

with energy. If we were to extend this formula to negative absolute temperatures, exponentially increasing distributions would result. Because the distribution needs to be normalized, at positive temperatures a lower bound in energy is required, as the probabilities P_i would diverge for $E_i \rightarrow -\infty$. Negative temperatures, on the other hand, demand an upper bound in energy ($E_i \rightarrow \infty$). In daily life, negative temperatures are absent, because kinetic energy in most systems, including particles in free space, only provides a lower energy bound. Even in lattice systems, where kinetic energy is split into distinct bands, implementing an upper energy bound for motional degrees of freedom is challenging, because potential and interaction energy need to be limited as well (3, 4). So far, negative temperatures have been realized in localized spin systems (5–7), where the finite, discrete spectrum naturally provides both lower and upper energy bounds. Here, we were able to realize a negative temperature state for motional degrees of freedom.

In Fig. 1A, we schematically show the relation between entropy S and energy E for a thermal system possessing both lower and upper energy bounds. Starting at minimum energy, where only the ground state is populated, an increase in energy leads to an occupation of a larger number of states and therefore an increase in entropy. As the temperature approaches infinity, all states become equally populated and the entropy reaches its maximum possible value S_{max} . However, the energy can be increased even further if high-energy states are more populated than low-energy ones. In this regime, the entropy decreases with energy, which, according to the thermodynamic definition of temperature (8) ($1/T = (\partial S/\partial E)$), results in negative temperatures. The temperature is discontinuous at maximum entropy, jumping from positive to negative infinity. This is a consequence of the historic definition of temperature. A continuous and monotonically increasing temperature scale would be given by $\beta = -1/k_B T$, also emphasizing that negative temperature states are hotter than positive temperature states, i.e., in thermal contact, heat would flow from a negative to a positive temperature system.

Because negative temperature systems can absorb entropy while releasing energy, they give rise to several counterintuitive effects, such as Carnot engines with an efficiency greater than unity (4). Through a stability analysis for thermodynamic equilibrium, we showed that negative temperature states of motional degrees of freedom necessarily possess negative pressure (9) and are thus of fundamental interest to the description of dark energy in cosmology, where negative pressure is required to account for the accelerating expansion of the universe (10).

Cold atoms in optical lattices are an ideal system to create negative temperature states because of the isolation from the environment and independent control of all relevant parameters (11). Bosonic atoms in the lowest band of a

Summary

